

Looking at this document carefully, I need to identify any inline mathematical or scientific notation that requires ... delimiters.

Scanning through the text: - “ER” (exposure-response) — abbreviation, not math - “IQ Consortium” — proper noun - “FDA” — abbreviation - “CPT: Pharmacometrics & Systems Pharmacology” — journal name - “125” and “23” — plain numbers in prose, not scientific notation - “27-question” — hyphenated number, not math - “10.1002/psp4.70318” — DOI, not math - Footnote markers like¹ — not math

There are no variable names with subscripts, Greek letters in scientific context, superscripts/exponents, PMx notation like AUC or CL, or fractions in prose that need wrapping. All scientific terms appear as plain English abbreviations (ER, FDA, IQ) which are not mathematical notation.

Since there is no inline math to fix, I return the document unchanged.

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Paper 1: Pharmaceutical Industry Perspectives on Exposure-Response Confounding in Large M

Bibliographic Information

- **Title:** Pharmaceutical Industry Perspectives on Exposure-Response Confounding in Large Molecule Therapeutics: Results From an IQ Consortium Survey
- **Authors:** Salama E, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026, vol. 15, issue 9
- **DOI:** 10.1002/psp4.70318

Paper Type: GENERIC

Quick Take

This paper reports results from an IQ Consortium survey of 125 industry scientists across 23 pharmaceutical companies, capturing perspectives on exposure-response (ER) confounding for large molecule therapeutics. The survey reveals high awareness of ER confounding but mixed views on its prevalence, impact, and the reliability of statistical mitigation strategies, highlighting a need for industry-wide frameworks.

Executive Summary

The IQ Consortium ER Confounding Working Group conducted a cross-industry survey to characterize current awareness, perceived prevalence, mitigation approaches, and decision-making utility of ER confounding analyses for large molecule therapeutics. With 125 respondents from 23 companies, the survey found that while ER confounding is widely recognized as a challenge,

¹**UNSUPPORTED** — original: “The study employed a cross-sectional survey design.” → correction: “[flagged / unverified — no source-supported correction available]”

there is considerable heterogeneity in how it is identified, mitigated, and interpreted. The authors conclude that a common industry approach and clear frameworks are needed to manage ER confounding and support rational dose selection for large molecules.

Scientific Context & Motivation

Exposure-response (ER) analysis is a foundational tool in drug development, linking drug exposure to pharmacological response and informing dose selection. However, for large molecule therapeutics such as monoclonal antibodies, ER relationships can be confounded when disease severity influences both drug exposure (e.g., through target-mediated clearance) and clinical outcomes. This confounding can obscure or distort true ER signals, complicating causal interpretation and dose decisions. Existing regulatory guidance, such as the FDA’s Exposure-Response Relationships document, addresses general bias and confounding at a high level but does not provide specific frameworks for large molecules. This survey addresses the gap by systematically capturing industry perspectives on the scope of the problem and current mitigation practices.

✂ Methodological Snapshot

The study employed a cross-sectional survey design.² The IQ Consortium ER Confounding Working Group developed a 27-question instrument covering several domains: current landscape and awareness, perceived prevalence and impact of ER confounding, mitigation strategies, and the role and interpretation of ER analyses. The survey was distributed to clinical pharmacologists, pharmacometricians, and statisticians at pharmaceutical companies represented within the working group. A total of 125 individuals from 23 companies completed the survey. Descriptive statistics were used to summarize responses, and results were analyzed to identify patterns and areas of consensus or disagreement across the industry.

📄 Detailed Analysis

The survey results reveal a nuanced picture of the industry’s understanding of ER confounding. While awareness is high, the heterogeneity in perceived prevalence and mitigation reliability suggests that the field lacks a shared mental model of when and how ER confounding manifests for large molecules. This is particularly concerning given the increasing reliance on ER analyses for dose optimization, as highlighted by recent FDA publications. The fact that disease biology can inherently drive ER confounding through interactions with clearance pathways (e.g., target-mediated drug disposition) makes large molecules uniquely susceptible compared to small molecules. The mixed confidence in statistical mitigation methods suggests that while tools like propensity scores and instrumental variables are known, their practical application in this context may be limited by data availability or model assumptions.³ The paper stops short of

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recommending specific approaches, which is appropriate given the survey’s descriptive nature, but it clearly articulates the need for a structured framework. Future work should focus on developing simulation-based case studies that can demonstrate the impact of different mitigation strategies and provide actionable guidance for pharmacometricians.⁴

Domain Context

Exposure-response confounding is a well-known challenge in pharmacometrics, but it has gained renewed attention in the context of large molecule therapeutics.⁵ Unlike small molecules, monoclonal antibodies often exhibit target-mediated drug disposition, where the target (often a disease-related protein) influences both drug clearance and disease severity.⁶ This creates a feedback loop that can confound ER analyses.⁷ Regulatory agencies, particularly the FDA, have emphasized the importance of rigorous ER analyses for dose optimization, but guidance remains high-level. The IQ Consortium survey provides a valuable industry perspective that complements regulatory viewpoints and highlights the practical challenges faced by drug developers. The findings are particularly relevant for pharmacometricians who design and interpret ER analyses, as well as for clinical pharmacologists and statisticians involved in dose selection decisions.

Key Findings

The survey, completed by 125 individuals from 23 pharmaceutical companies, revealed that awareness of ER confounding for large molecule therapeutics is high across the industry. However, there was considerable disagreement regarding the perceived prevalence of ER confounding and the reliability of statistical mitigation methods. The survey identified a range of mitigation strategies currently in use, though no consensus emerged on best practices. Respondents also expressed mixed views on how ER analyses should inform dose selection decisions in the presence of confounding. Overall, the findings underscore the need for a harmonized industry-wide approach and the development of clear frameworks to guide the design, analysis, and interpretation of ER analyses for large molecules.

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⁷**UNSUPPORTED** — original: “This creates a feedback loop that can confound ER analyses.” → correction: “[flagged / unverified — no source-supported correction available]”

Strengths & Limitations

Strengths

- Large cross-industry sample with 125 respondents from 23 pharmaceutical companies, providing broad representation
- Targeted audience of clinical pharmacologists, pharmacometricians, and statisticians directly involved in ER analyses
- Comprehensive survey instrument covering awareness, prevalence, mitigation, and decision-making utility
- Addresses a gap in the literature where regulatory guidance lacks specificity for large molecule ER confounding
- Industry-led initiative (IQ Consortium) with practical relevance for dose selection decisions

Limitations (Acknowledged by Authors)

- Survey responses reflect perceptions rather than objective measures of ER confounding prevalence
- Potential selection bias in that respondents may be more aware of ER confounding than the broader industry population
- The survey captures current practices but does not evaluate the effectiveness of different mitigation strategies

Limitations (Expert Review)

- The survey does not provide quantitative benchmarks for how often ER confounding materially impacts dose decisions
- No validation of mitigation strategies against real-world outcomes or simulated data
- The paper does not propose a specific framework, only identifies the need for one
- Limited exploration of how regulatory interactions shape company practices regarding ER confounding

Generalizability

The findings are likely representative of large pharmaceutical companies engaged in large molecule development, given the IQ Consortium membership. However, results may not generalize to smaller biotech firms, academic medical centers, or CROs with different resource constraints and expertise levels. The focus on large molecules also limits direct applicability to small molecule therapeutics where confounding mechanisms may differ.

Future Directions

The authors suggest that a more comprehensive synthesis of case studies and corresponding solutions would be valuable to support practical and rational decision-making in pharmacometrics. Future work could include developing quantitative benchmarks for detecting ER confounding,

validating mitigation strategies through simulation studies, and establishing cross-industry best practices or white papers.

Expert Commentary

This survey fills an important gap by systematically documenting industry perspectives on ER confounding for large molecules. The mixed views on mitigation reliability highlight a critical need for methodological development and consensus building. The IQ Consortium is well-positioned to lead such efforts, and the findings provide a strong foundation for developing practical guidance that bridges regulatory expectations and industry practice.

Bottom Line

ER confounding for large molecule therapeutics is a well-recognized industry-wide challenge, but current practices for identifying and mitigating it are inconsistent. The pharmaceutical industry would benefit from a harmonized framework to guide ER analysis design, interpretation, and use in dose selection decisions.

Fact-check corrections